A Euclidean-stratification upper bound for Heisenberg-cylinder discrepancy

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Status. Candidate partial result, likely valid. The theorem improves the known upper exponent in a natural Lebesgue-measure subcase; it does not determine the sharp exponent.

1 Source question

Brandolini, Monguzzi, and Monti [1] consider $\mathbb{H}^n = \mathbb{C}^n \times \mathbb{R}$, with group law

$$(z,t) \circ (w,s) = \left(z+w, t+s+\tfrac{1}{2}\operatorname{Im}(z \cdot \overline{w})\right),$$

homogeneous dimension $Q = 2n+2$, and cylinders

$$B_\rho = \{(z,t) : |z| \le \rho,\ |t| \le \rho^2\}.$$

For a probability measure μ and points $p_1, \dots, p_N$, they define

$$D_N(g;\rho) = \sum_{j=1}^{N} \mathbf{1}_{g \circ B_\rho}(p_j) - N\mu(g \circ B_\rho).$$

Their Theorem 2 proves

$$\int_0^1 \int_{\mathbb{H}^n} |D_N(g;\rho)|^2 \, dg \, d\rho \gtrsim N^{1-1/(Q-2)}$$

for upper Ahlfors-regular μ. They ask whether this is sharp and record, for suitable normalized Lebesgue measures, the general metric-space upper bound $N^{1-1/Q}$.

Figure 1 is the question on page 3 of the source paper.

The proof of the theorem combines classical arguments of Roth and Montgomery with the harmonic analysis on the Heisenberg group, in particular with the use of the group Fourier transform defined in terms of the Schrödinger representations of $\mathbb{H}^n$.

Let us clarify why cylindrically symmetric sets arise naturally in our analysis. The group Fourier transform maps a function in $L^1(\mathbb{H}^n)$ to a family of operators on $L^2(\mathbb{R}^n)$. For cylindrically radial functions, these operators have a particularly convenient form: they are diagonal in the Hermite basis. This motivates defining the discrepancy with respect to the sets B_ρ, which form a simple, explicit family of cylindrically radial sets. Moreover, this choice allows us to exploit an explicit description of the Fourier transform of cylindrically radial functions on the Heisenberg group; see Section 3.

From a geometric point of view, it would also be natural to consider discrepancy with respect to Heisenberg balls associated with the Korányi metric. The corresponding L^2-discrepancy problem, however, leads to a substantially different Fourier-analytic setting and is not addressed in this work. In the same circle of ideas, we point out to the reader the papers [20, 21].

It is natural to ask whether Theorem 2 is sharp. In Section 5 we apply a general upper bound for the discrepancy of Ahlfors regular measures on metric measure spaces to our setting, which yields $N^{1-\frac{1}{Q}}$. At present, we do not know which rate is the correct one. The available results in the metric measure space setting are quite general and do not incorporate the geometry of the families of test sets used to evaluate the discrepancy; rather, they only depend on the dimension of their boundaries. Moreover, examples on the torus suggest that when the boundaries of the test sets exhibit singularities, the discrepancy can be smaller than what one would otherwise expect, see [11] and [12]. This latter observation supports the conjecture that our lower bound is the optimal one.

The rest of the paper is organized as follows. Sections 2 and 3 collect the basic material on the Heisenberg group and recall the harmonic analysis tools used throughout the paper. In Section 4 we prove our main result (Theorem 2); its proof relies on two main estimates that will be proved in Sections 7 and 8. The preliminary technical ingredients needed to prove such estimates are provided in Section 6.

In the entire paper we adopt the following notation conventions. Given two quantities A and B that depend on some parameter x, in the following we will write $A \approx B$ meaning that there are two positive constants $c, C > 0$ independent of the relevant values of the parameter x such that

$$c\,B \leqslant A \leqslant C\,B.$$

If one can take $c = C$, we also write $A \simeq B$. Moreover, we write $A \lesssim B$ (resp. $A \gtrsim B$) to denote a one-sided inequality $A \leqslant C\,B$ (resp. $A \geqslant c\,B$) for some constant $C > 0$ (resp. $c > 0$).

We also do not attempt to keep track of constants: a positive constant, usually denoted by c, may change from line to line. Finally, whenever we are dealing with an integration on $\mathbb{R}^n$ or $\mathbb{H}^n$, unless specified, it is understood that the underlying measure is the Lebesgue measure.

We conclude the introduction by thanking Alessio Martini for stimulating discussions and valuable suggestions concerning the heat kernel on the Heisenberg group.

2. The group Fourier transform

In this section we recall the definition of the group Fourier transform, its main properties and we state the Plancherel identity. For further details, we refer the reader to the standard references [18, 30].

2.1. **The Schrödinger representation of $\mathbb{H}^n$ and the group Fourier transform.** Here and in the subsequent subsections we recall some facts about the group Fourier transform on $\mathbb{H}^n$. For details we refer the reader to the standard references [18], [22], and [30].

Identifying $z = x + iy$ we can also describe $\mathbb{H}^n$ as the space $\mathbb{R}^{2n+1}$ endowed with the group operation

$$(x_1, y_1, t_1) \circ (x_2, y_2, t_2) = \left(x_1 + x_2, y_1 + y_2, t_1 + t_2 + \frac{1}{2}(y_1 x_2 - x_1 y_2)\right).$$

Let $\lambda \neq 0$, and consider the map π_λ from $\mathbb{H}^n$ to the space of bounded linear operators on $L^2(\mathbb{R}^n)$, defined by

$$\pi_\lambda(z,t)\varphi(\xi) = \pi_\lambda(x,y,t)\varphi(\xi) = e^{i\lambda\left(x\cdot\xi + \frac{1}{2}x\cdot y + t\right)}\varphi(\xi + y),$$

for $(z,t) = (x,y,t) \in \mathbb{H}^n$ and $\varphi \in L^2(\mathbb{R}^n)$. This defines a unitary representation of the Heisenberg group $\mathbb{H}^n$ on the Hilbert space $L^2(\mathbb{R}^n)$, known as the *Schrödinger representation*. Observe that setting

Figure 1: The sharpness question in arXiv:2601.15850, page 3.

2 Proof intuition

The source upper bound partitions the support using the intrinsic Heisenberg metric, whose dimension is Q. For Lebesgue measure, however, the ambient Euclidean coordinate dimension is only

$$d = 2n + 1 = Q - 1.$$

Partition a cube into equal-volume cells of Euclidean diameter $h \asymp N^{-1/d}$, and choose one random point in each cell. For a fixed test cylinder, cells entirely inside or outside make no random error; only cells crossing its boundary contribute variance. A relevant Heisenberg translate is an affine shear of a Euclidean ball times an interval. On the bounded parameter region that can interact with the cube, these shears have uniformly bounded condition number, and their boundary layers have Euclidean volume $O(h)$. Thus only $O(Nh)$ cells contribute. Averaging over all translations and radii and then choosing one realization gives the result.

The remaining gap is real: this argument sees ordinary Euclidean codimension one, not the effective horizontal dimension $Q - 2$ suggested by the lower bound.

3 The partial theorem

Let $E = [-1/2, 1/2]^d \subset \mathbb{R}^d = \mathbb{H}^n$ and let μ be Lebesgue measure restricted to E, so $\mu(E) = 1$.

Theorem 1. *For every $N \geq 2$, there exist points $p_1, \dots, p_N \in E$ such that*

$$\int_0^1 \int_{\mathbb{H}^n} |D_N(g; \rho)|^2 \, dg \, d\rho \leq C_{n,E} N^{1-1/(Q-1)}.$$

Consequently, in this Lebesgue-measure subcase,

$$c_{n,E} N^{1-1/(Q-2)} \leq \inf_{p_1,\dots,p_N \in E} \int_0^1 \int_{\mathbb{H}^n} |D_N(g; \rho)|^2 \, dg \, d\rho \leq C_{n,E} N^{1-1/(Q-1)}.$$

In particular, the upper exponent $1 - 1/Q$ recorded in the source improves to $1 - 1/(Q - 1)$ for this subcase.

We use two elementary lemmas.

Lemma 1 (Equal-volume Euclidean cells)**.** *For every $d \geq 1$ and $N \geq 1$, the unit cube admits a measurable partition $E = E_1 \sqcup \cdots \sqcup E_N$ such that*

$$|E_j| = N^{-1}, \qquad \mathrm{diam}_{\mathrm{Euc}}(E_j) \leq C_d N^{-1/d}.$$

Proof. This follows by induction on d. For $d = 1$, use N equal intervals. For the inductive step, choose $m \asymp N^{1/d}$ and write $N = N_1 + \cdots + N_m$, where all positive integers N_j are comparable to N/m. Divide the first coordinate into slabs of widths N_j/N. In the j-th slab, apply the $(d - 1)$-dimensional construction to its unit cross-section with N_j cells. Every resulting box has volume $1/N$, first-coordinate width $O(N^{-1/d})$, and transverse diameter $O(N_j^{-1/(d-1)}) = O(N^{-1/d})$. □

Lemma 2 (Uniform Euclidean boundary layers)**.** *There is a compact set $K \subset \mathbb{H}^n$ and a constant C such that the following holds for $0 < h \leq 1$, $0 \leq \rho \leq 1$, and $g \in \mathbb{H}^n$. If the Euclidean $2h$-neighborhood of $\partial(g \circ B_\rho)$ meets E, then $g \in K$, and*

$$|E \cap \{x : \mathrm{dist}_{\mathrm{Euc}}(x, \partial(g \circ B_\rho)) \leq 2h\}| \leq Ch.$$

Proof. Write $g = (z, t)$. In Euclidean coordinates, left translation is the affine map

$$T_g(w, s) = \left(z + w, t + s + \tfrac{1}{2}\operatorname{Im}(z \cdot \overline{w})\right).$$

It has determinant one. If a translated cylinder boundary comes within distance two of the fixed cube, then $|z|$ is bounded because its horizontal projection is the ball $z + \{|w| \le \rho\}$, with $\rho \le 1$. The central coordinate t is then bounded as well, since $|s| \le \rho^2$ and the shear term is bounded. This gives a fixed compact parameter set K. For $g \in K$, the linear parts of T_g and T_g^{-1} have uniformly bounded operator norms.

Before applying T_g, the set B_ρ is the Euclidean product of a $2n$-ball of radius ρ and an interval of length $2\rho^2$. If $\rho \ge h$, its two cap layers and its lateral layer have total volume at most $C_n h$, including the edge neighborhoods. If $\rho < h$, the whole $O(h)$-neighborhood has volume

$$O\big((\rho + h)^{2n}(\rho^2 + h)\big) = O(h^{2n+1}) = O(h).$$

The uniformly bounded affine maps T_g preserve this estimate up to a constant. Intersecting with E can only decrease volume. $\square$

Proof of Theorem 1. Take the partition from Lemma 1, and put $h = C_d N^{-1/d}$. The finitely many N for which $h > 1$ are absorbed by enlarging the final constant, so assume $h \le 1$. Independently in each E_j, choose X_j according to normalized Lebesgue measure on that cell. For a fixed measurable test set $R = g \circ B_\rho$, let

$$p_j = N|E_j \cap R|, \qquad Y_j = \mathbf{1}_R(X_j) - p_j.$$

Then $\mathbb{E}Y_j = 0$, the variables are independent, and

$$\mathbb{E}|D_N(g;\rho)|^2 = \sum_{j=1}^{N} p_j(1 - p_j). \tag{1}$$

If p_j is zero or one, its summand vanishes. Otherwise the cell meets both R and its complement. Since its diameter is at most h, the whole cell lies in the Euclidean $2h$-neighborhood of ∂R. Lemma 2 and the equal cell volumes therefore give

$$\#\{j : 0 < p_j < 1\}\, N^{-1} \le Ch,$$

and hence, by (1),

$$\mathbb{E}|D_N(g;\rho)|^2 \le CNh. \tag{2}$$

Moreover, a nonzero summand in (1) forces the boundary layer to meet E, so Lemma 2 restricts g to the fixed compact set K. Tonelli's theorem and (2) yield

$$\mathbb{E}\int_0^1 \int_{\mathbb{H}^n} |D_N(g;\rho)|^2 \, dg \, d\rho \le C|K|Nh \le C' N^{1-1/d}.$$

Some realization of $X_1, \ldots, X_N$ is no larger than this expectation. Since $d = Q - 1$, this is the asserted upper bound.

Finally, Lebesgue measure on E is upper Ahlfors regular for the Korányi metric because a Korányi ball of radius r has Lebesgue volume $c_n r^Q$. The source's Theorem 2 therefore supplies the displayed lower bound. $\square$

4 Verification, limitations, and novelty status

The proof has no numerical dependency. It explicitly handles the two potential uniformity issues: small radii in Lemma 2, and the unbounded translation domain via the compact active set K.

The result is partial in two ways. First, it treats normalized Lebesgue measure on a cube (the same natural type of measure used for the source upper bound), not every upper Ahlfors-regular probability measure. Second, it leaves the exponent gap

$$1-\frac{1}{Q-2} \quad \text{versus} \quad 1-\frac{1}{Q-1}.$$

An anisotropic cell count cannot close this gap because the sheared flat caps vary by first order in both horizontal and central cell widths.

A bounded search covered the exact title and sharpness language, the source's references on metric discrepancy and halfspaces, the phrases "Heisenberg cylinder discrepancy" and "quadratic discrepancy Heisenberg sharp", and the June 2026 paper on multidimensional nonsmooth convex bodies [3]. No exact statement of the $Q-1$ upper bound for this Heisenberg family was found through 11 August 2026. The method is nevertheless close to the standard stratified-sampling argument of Brandolini et al. [2], so novelty confidence is moderate rather than high.

Human-review recommendation. Review the uniform affine-shear boundary-layer estimate and confirm that the source's intended comparison includes the Lebesgue cube subcase. If valid, the packet should be retained as a substantial partial upper-bound improvement, not as a resolution of the conjectured sharp exponent.

References

[1] L. Brandolini, A. Monguzzi, and M. Monti, *Quadratic discrepancy estimates for probability measures on the Heisenberg group*, arXiv:2601.15850 (2026).

[2] L. Brandolini, W. W. L. Chen, L. Colzani, G. Gigante, and G. Travaglini, *Discrepancy and numerical integration on metric measure spaces*, J. Geom. Anal. 29 (2019), 328–369.

[3] R. Bramati, L. Brandolini, and A. Monguzzi, *Discrepancy estimates for multi-dimensional non-smooth convex bodies: a case study*, arXiv:2606.07028 (2026).